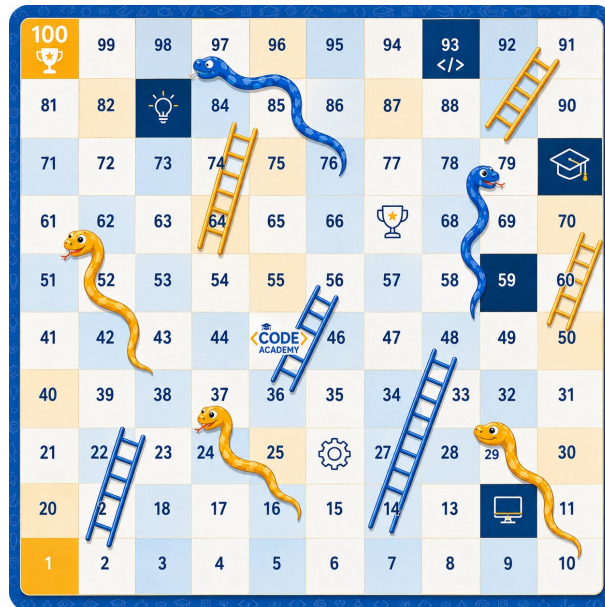


Snake and Ladder 1

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

taketaketake is playing a simplified Snake and Stairs game.



The board contains cells numbered from 1 to 100, and **taketaketake** starts at cell 1. Some cells contain a snake or a stair. Each is represented by two integers x and y :

- If $x < y$, landing on cell x moves **taketaketake** to cell y (stair).
- If $x > y$, landing on cell x moves **taketaketake** to cell y (snake).

For each dice roll:

- **taketaketake** moves forward by the dice value.
- If he lands on a snake or stair, he immediately moves to its destination.
- If the move goes beyond cell 100, it is ignored.
- If **taketaketake** reaches cell 100, the game ends immediately.

Given the snakes, stairs, and dice rolls, determine **taketaketake**'s final position.

Important Note: After landing on a snake or stair, the player moves to its destination once. No additional snake or stair is applied during the same dice roll.

Input

The first line contains two integers m ($1 \leq m, q \leq 30$), the number of snakes and stairs.

The next m lines contain two integers each line x and y ($1 \leq x, y \leq 100$).

The last line contains q dice rolls $d_1, d_2, \dots, d_n$ ($1 \leq d \leq 50$)

Output

Output a single integer. The final position that **taketaketake** lands on.

Examples

standard input	standard output
1 1 3 4 1	2
4 3 2 3 4 7 3 9 10 3 1 7 1	7